Personal Psychology 20

Unit 7

nextSTEP
SENIOR HIGH SCHOOL
Fort Saskatchewan 780 992 0101
Sherwood Park 780 464 1899
Vegreville 780 632 7998

STUDENT NAME:

Value
22

Part One: Multiple-Choice Items

Read each item carefully and decide which choice best completes the statement or answers the question. Write the letter of your choice in the blank before the item.

Point of clarification regarding sensation and perception:

Sensation refers to the process of receiving stimuli from the environment using one of the five senses and then having that information forwarded to our brains for processing. Perception is the processing and interpretation of the sensory stimuli we receive to better understand our environment.

Use the following information to answer multiple-choice items 1 and 2.

Robert loves vegetables and after he eats his bowl of steamed broccoli with fresh butter and sea salt, he decides that broccoli is his new favourite vegetable.

_____ 1. As Robert places a broccoli floret in his mouth, breathes in deeply, and begins to chew, he is experiencing a(n)

A. illusion
B. sensation
C. perception
D. hallucination

_____ 2. When Robert states that broccoli is his new favourite vegetable, this is due to his

A. illusion
B. sensation
C. perception
D. hallucination

_____ 3. ESP is

A. the sixth sense
B. a visual illusion
C. not scientifically proven
D. something everyone has

_____ 4. Eyes are designed to detect

A. colour
B. light waves
C. features of objects
D. electrical impulses

_____ 5. The pupil

A. expands in dim light
B. contracts in dim light
C. enlarges in bright light
D. does not change shape

_____ 6. The coloured part of the eye is called the

A. iris
B. lens
C. pupil
D. cornea

_____ 7. The dark centre of the eye is called the

A. iris
B. lens
C. pupil
D. cornea

_____ 8. The transparent covering at the front of the eye is called the

A. iris
B. lens
C. pupil
D. cornea

_____ 9. This part of the eye controls the size of the pupil

A. iris
B. lens
C. cornea
D. ciliary muscles

_____ 10. This part of the eye focuses the images we see

A. iris
B. lens
C. pupil
D. cornea

_____ 11. One colour is distinguished from another colour by its

A. power
B. intensity
C. wavelength
D. feature detectors

_____ 12. Specialized cells called _____ are concentrated mainly in the centre of the retina and work _____.

A. rods; in dim light
B. cones; in dim light
C. rods; to detect colour
D. cones; to detect colour

_____ 13. Specialized cells called _____ are concentrated mainly around the edge of the retina and work _____.

A. rods; in dim light
B. cones; in dim light
C. rods; to detect colour
D. cones; to detect colour

_____ 14. The blind spot is caused by

A. the iris closing
B. colour blindness
C. a concentration of photoreceptors
D. the space where the optic nerve leaves the eye

Point of clarification regarding cellular layers in the retina:

Page 113 incorrectly states that light first goes through the photoreceptor layer (rods and cones) then the bipolar layer and finally the ganglion cell layer. What actually happens is that light entering the eye passes through the ganglion cell layer then the plexiform and nuclear layers (with bipolar cells) on its way to the photoreceptor layer where the rods and cones are located. The rods and cones then activate and send *signals* **back** to the ganglion cells through the bipolar cells.

_____ 15. Individuals with monochromatism see only in _____ because their _____ do not function.

A. the three primary colours; rods
B. the three primary colours; cones
C. shades of white, grey, and black; rods
D. shades of white, grey, and black; cones

_____ 16. Individuals with protanopia perceive only shades of

A. red and blue
B. red and yellow
C. blue and yellow
D. white and black

_____ 17. Which set of colours are complementary?

A. red and green
B. blue and green
C. yellow and blue
D. purple and orange

_____ 18. When you mix two primary colours you get a

A. tint
B. shade
C. secondary colour
D. intermediate colour

_____ 19. The cochlea helps to transmit vibration into neural impulses by

A. detecting features
B. absorbing sound waves
C. stimulating the auditory nerve
D. causing hairs on its surface to vibrate

_____ 20. Pheromones

A. are hormones
B. have strong odours
C. signal sexual receptivity
D. are only active in insects

_____ 21. The placebo effect occurs when people report feeling

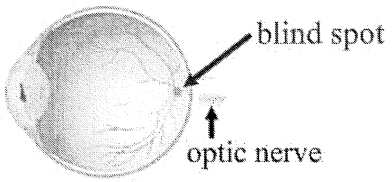
A. less pain when they have taken real medication
B. more pain when they have taken real medication
C. less pain when they have taken inactive (fake) medication
D. more pain when they have taken inactive (fake) medication

_____ 22. In the wild, _____ is considered the most important sense.

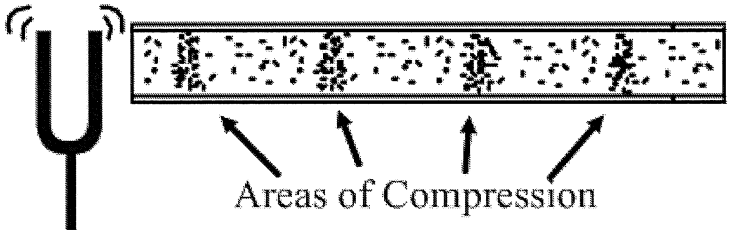
A. smell
B. touch
C. vision
D. hearing

Corrections to Course Content Regarding the Five Senses

- **Sight:** The diagram on page 113 incorrectly indicates that the blind spot is located in the vitreous humour at the back of the eye just in front of the retina. The blind spot is actually located where the optic nerve enters/exits the retina.



- **Sound:** The diagram on page 125 depicts sound waves as transverse waves (like light waves) with high and low points (crests and troughs) in the waves. Sound, in fact, is a longitudinal wave. A correct diagram of longitudinal waves formed when a tuning fork is struck near a pipe with air in it is as follows:



  - The distance between two consecutive areas of compression (wave cycles) determines the pitch of the sound wave. Long distances between compressions result a lower pitch (lower frequency). Shorter distances between compression result in a higher pitch (higher frequency). *Frequency* is the number of compressions (wave cycles) in a given time period.

  - The number of air molecules disturbed or compacted in a compression area relates to the sound pressure of a sound wave. The pressure (also known as *amplitude*) relates to the loudness or quietness of a sound. Without molecules (air, water, etc.) to transmit the energy (pressure) of a sound wave, there is no audible sound. In other words, if a hammer strikes a metal bar in a container that has had all the air removed (i.e., a vacuum), you would not hear anything.

- **Taste & Smell:** The course materials correctly identify that taste and smell are intimately entwined and function together to enhance the experience of eating. On page 126, however, the materials identify four basic tastes: sour, salty, sweet, and bitter. There is actually a fifth basic taste called umami. The word umami is borrowed from the Japanese culture and is translated as "pleasant savory taste." Ripe tomatoes are thought to be rich in umami flavour!

- **Touch:** This sense includes pain, temperature (hot and cold), as well as pressure. There are two typographical errors on page 128 that substitute "pleasure" for "pressure." Please note the correct term is pressure.

Value
10

Part Two: Matching – The Auditory System

Choose the letter of the term that best describes the concept or completes the sentence. Terms may not be used more than once.

Statements

_____ 1. This is term refers to the strength, pressure, or "loudness" of a sound wave.

_____ 2. This is the outer shell of the ear.

_____ 3. This structure transfers auditory information to the brain.

_____ 4. This curled tube-like structure contains fluid.

_____ 5. This is the first of three tiny bones in the inner ear that relay vibrations from the external environment.

_____ 6. This refers to the number of compressions in a given time period.

_____ 7. Vibrations from the hammer are relayed to this tiny bone in the inner ear.

_____ 8. This area receives and translates auditory information.

_____ 9. This blocks and protects the delicate bones of the inner ear from the external world.

____ 10. This structure transfers vibrations to the cochlea.

Terms

A. amplitude
B. anvil
C. auditory cortex
D. auditory nerve
E. cochlea
F. frequency
G. hammer
H. pinna
I. stirrup
J. tympanic membrane

Value 10

Part Three: True or False

Read each item carefully and decide if the statement is true or false. Write **T** or **F** in the blanks to indicate your choices.

_____ 1. Colour blindness is genetic.

_____ 2. All animals have colour vision because colour is used to attract mates.

_____ 3. A tertiary colour is a combination of a primary colour and a secondary colour.

_____ 4. People with hearing loss may not realize their hearing has deteriorated.

_____ 5. Taste buds are only located on the edges of the tongue but taste buds respond to all tastes to varying degrees.

_____ 6. People with brain injuries that result in a loss of smell may then also experience a loss of taste.

_____ 7. Some parts of our skin have more sensory touch receptors than other parts of our skin.

_____ 8. Water droplets can act like prisms and separate light into many colours.

_____ 9. Humans attend to sound only in one ear.

____ 10. When you add black to a colour you get a tint.

Value
25

Part Four: Written-Response Items

Respond to the following items on the blanks below the items.

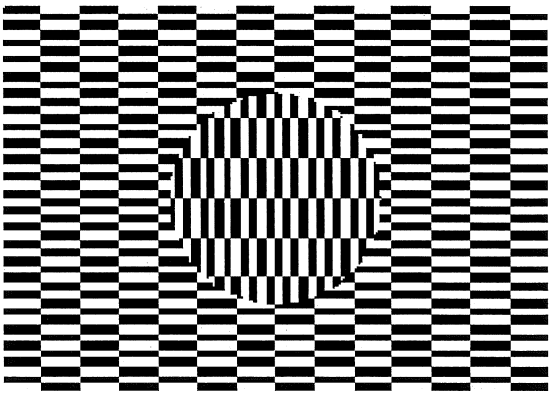
Addition Information Regarding Optical Illusions

Mechanisms that create optical illusions are located in various places along the visual pathways. Some of these mechanisms come from the retina itself, but most result from the way that the images captured with the eyes are reconstructed by the visual cortex.

The information our brain receives from our eyes doesn't always accurately represent reality. For example, some of the information received from the 125 million photoreceptors is lost as the information flows to 100 times fewer ganglion cells before being transmitted to the brain by the optic nerve.

To compensate for the loss of information, the brain introduces abstract factors that complete or amplify the components of reality that are transmitted. In doing so, the brain sometimes creates an impression of unity/logic where there is none—in other words, an optical illusion. There are many types of optical illusions including those related to depth, colour, motion, and size. The written-response items 1 and 2 relate to optical illusions.

2 1. Below is an optical illusion. Hold your head still and move this page to the left then right and describe what you see.



_____

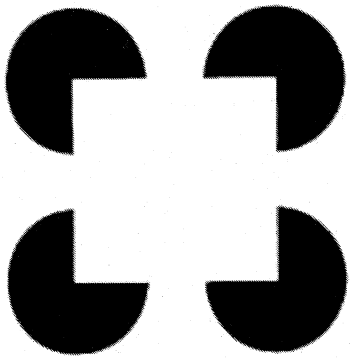
_____

_____

_____

_____

2 2. Describe what you see in the optical illusion below.



1 3. What is the name of the taste sensation (basic taste) with a Japanese name?

2 4. What happens if you plug your nose and eat an apple?

2 5. How does the mind compensate when experiencing sensory deprivation?

_____

_____

_____

_____

_____

2 6. How does the mind compensate when experiencing sensory overload?

_____

_____

_____

_____

_____

14 7. Select **one** of the following two options on the following pages and complete the associated tasks. Remember to enumerate (number) your responses on the response page for Option A so your marker knows the item to which you are responding. Responses for Option B may be provided within the Option itself.

Option A: Hot and Cold

As you have learned, the cutaneous sense (touch) involves pressure, pain, and temperature. For this option you may need to conduct internet (or other) research and you will conduct a short experiment regarding temperature. You will need three tall glasses (identical glasses if possible), water of varying temperatures, and a clock/watch/timer to time yourself. Once you have the materials, complete the following steps in the order they appear then answer the written-response items that follow in the space provided on page 14:

Steps:

- Fill one glass with ice water (very cold), fill one glass with room-temperature water (you can leave the glass of water on a table for a few hours to reach room temperature), and fill one glass with very hot water (not boiling).
- Grasp the glass of hot water with one hand, making sure the palm of your hand is touching the glass.
- Grasp the glass of ice water with your other hand, making sure the palm of your hand is touching the glass.
- Hold the glasses (hot water in one hand, ice water in the other) for 60 seconds.
- After 60 seconds, grasp the room-temperature glass with both hands, palms touching the glass.

1. Describe, with some detail, the sensations of touching the room-temperature glass after holding the other two glasses for 60 seconds. (4 marks)

2. What is the commonly used temperature scale in Canada called? (1 mark)

3. What is the commonly used temperature scale in the United States called? (1 mark)

4. What is the average body temperature of an adult in the temperature scale used in Canada? (2 marks)

5. Assessing a patient's temperature is part of a full physical examination. Besides taking a rectal (anal) reading of temperature, list two other sites that may be used for measuring temperature. (2 marks)

6. Body temperature fluctuates slightly. For a person who sleeps at night and stays awake during the day (no night/shift workers), at what time of day is body temperature the lowest? Be specific for full marks. (2 marks)

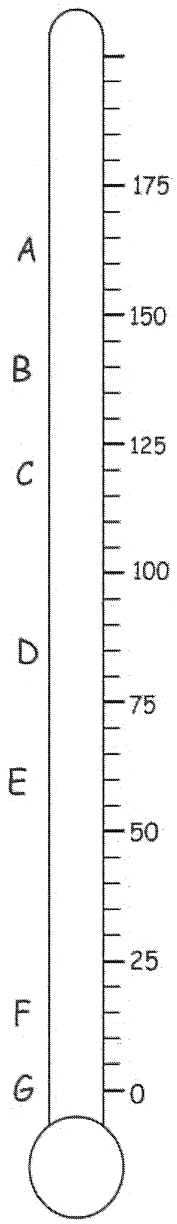
7. At what time of day is body temperature highest for a person who sleeps at night and stays awake during the day? Be specific for full marks. (2 marks)

---------- **or** ----------

Option B: Do You Hear what I Hear?

There are different types of hearing loss. Hearing loss may result from a tumor, from trauma to the ear, or even from noise. Exposure to noise is often measured in decibels. Conduct research (internet or otherwise), regarding noise produced from common activities and then complete the tasks that follow. You may respond on this page.

Sound levels in decibels (dB)



1. Above what decibel level is sound considered dangerous? (2 marks)

   _____

2. The thermometer diagram on the left goes from 0 decibels to over 175 decibels. Find three coloured pencils (red, yellow, green) or three highlighters (pink, yellow, green), and show, by colouring the thermometer, the range of decibels that is safe (green), the range of decibels that is borderline (yellow), and the range of decibels that is considered dangerous (red/pink). (3 marks)

3. On the left side of the thermometer are letters A-G. Correctly match the letter to the sound it represents. (7 marks)

   _____ Rock concert or ambulance
   _____ 12-gage shotgun
   _____ Busy city traffic
   _____ Whisper
   _____ Softest sound heard
   _____ Typical speech
   _____ Fireworks/firecrackers

4. What is tinnitus? (2 marks)

   _____
   _____
   _____
   _____
   _____
   _____

(Response Area for Option A.)

End of Assignment Booklet 7
After you have completed all items, submit your work to your marker.